

RISP: Retention and Invariance in Decision-Focused Strategy Pools for Non-Stationary Markets

An Explanatory Companion: Architecture, Mechanisms, Demonstrations, and Applications

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Abstract

Trading systems that must serve many market regimes with capacity-limited strategy specialists face two allocation problems that are usually conflated into one. The first is *between regimes*: market states recur on irregular schedules with long dormancy, and any mechanism that allocates capital, attention, or recalibration by chasing current reward—a trailing-Sharpe capital allocator, a learned mixture-of-experts gate—systematically starves and overwrites the specialist for whichever regime is dormant, then pays a full relearning cost in the very window where decisions matter most: the first days after the regime returns. The second is *within* a regime: each recurrence is a distinct draw from the regime’s environment distribution—the 2008 crisis is not 2020 is not 2022—so even a perfectly retained specialist fails out-of-distribution on the next episode if it was trained by empirical risk minimization on the last one. To address both, we present **RISP** (Regime-Invariant Specialist Pools), a population-learning framework for decision-focused (predict-then-optimize) trading built from three tightly coupled components: *per-regime decision heads* (linear SPO+ predictors feeding an exact portfolio optimizer, scored by decision regret rather than prediction error), *niche-affinity tracking* (the canonical exponentiated-gradient update on the regime simplex, driven by batched winner-take-all competition), and *episode-invariant niche training* (each specialist minimizes the variance plus mean of its regret surrogate across the historical episodes of its own regime). The converged regime-to-specialist assignment is *reward-independent*—a crisis specialist holds its niche by identity and idles through calm markets, retaining both its parameters and its episode memory—while the invariance objective makes what is retained *general* across episodes rather than calibrated to the last one. We prove a post-reactivation regret bound that splits into a forgetting deficit (zero for any reward-independent covering assignment; tending to the full relearning cost for any reward-chasing one) and an invariance gap (controlled by the episode-variance penalty, with a one-step Pinsker bridge from divergence-style invariance guarantees to regret). In controlled regime-switching markets at realistic signal-to-noise, every pre-registered prediction of the resulting 2×2 holds (20 seeds, Holm-corrected): the retention axis is worth 6.1% of post-reactivation regret ($p=1.3 \times 10^{-3}$), the invariance axis a further 4.4% ($p=6.1 \times 10^{-3}$), jointly 10.3% ($p=9.2 \times 10^{-7}$; 38% of the excess above the irreducible noise floor)—and the interaction is *super-additive*: the invariance objective is statistically inert inside a reward-driven monolith ($p=0.88$), because no training objective can help a head that was evicted during dormancy. The full system is statistically indistinguishable from a hand-pinned oracle assignment trained with the same objective ($p=0.72$): competition recovers the skyline it was never shown. We are equally explicit about what did *not* work: a pre-registered structure diagnostic on real crypto and commodity panels finds *no exploitable regime structure on the decision metric* under causal labelings, the real-data dissociation is therefore correctly null, our

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batching expectation for the competition window was refuted by the ablation, and the mechanism’s advantage *inverts* (up to -49%) when within-regime signal becomes strong enough to chase—a measured crossover, not a caveat. We close with an evidence-tiered application survey: multi-strategy capital allocation and risk-model committees (mechanism-level evidence), and the conditions under which one should *not* deploy the framework. Code and per-figure result files are released with the companion paper.

Keywords: regime switching; decision-focused learning; catastrophic forgetting; invariant risk; strategy pools; reward-independent assignment; competitive exclusion

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1 Introduction

“The time to repair the roof is when the sun is shining.”

—J. F. Kennedy, on a different allocation problem

Every multi-strategy trading operation maintains expertise it is not currently using. The crisis playbook idles through years of calm. The short-volatility book hibernates through stress. The breakout system waits out the range. This standing inventory of dormant competence is expensive —infrastructure, attention, calibration—and so every such operation also runs an allocator that decides, day by day, which strategies receive capital and which receive maintenance. The allocator that practice and machine learning both reach for first is *reward-chasing*: trailing performance weights, a learned gate, a router trained on realized P&L.

The GAUSE study [1] isolated, in controlled prediction domains, why that default quietly destroys the inventory it is supposed to manage: **a dormant regime emits no reward**. Whatever signal the allocator uses to protect capacity is exactly the signal a dormant regime does not produce, so reward-driven allocation reallocates the dormant specialist’s capacity to active regimes—hard eviction or gradual interference, the result is the same—and the system relearns its crisis playbook *during the crisis*. GAUSE’s resolution was structural: winner-take-all competition over niches converges to an assignment that is *independent of current reward*, so dormant specialists idle and retain, with no freezing schedule and no gate to train.

This companion document explains RISP, which ports that mechanism to decision-focused financial learning and discovers, in doing so, that retention is only *half* of the financial problem. The other half has no analogue in GAUSE’s setting and comes from the invariant-learning literature [2, 3, 4]: **regimes do not return unchanged**. A retained crisis specialist knows the *last* crisis—its correlations, its liquidity pattern, its episode-specific accidents. If it was trained by empirical risk minimization (ERM), it has loaded on those accidents (they were predictive, in-sample), and the next crisis, a fresh draw from the crisis distribution, punishes exactly that loading. Protecting the specialist’s memory protects the overfitting too. What must be retained is not the calibration to the last episode but the *decision-relevant structure common to all episodes*—and that is a property of the training objective, not of the allocator.

RISP therefore couples two mechanisms, each useless without the other:

1. **Reward-independent capacity assignment** (solves between-regime forgetting). Specialists acquire regimes through batched winner-take-all competition; an exponentiated-gradient affinity update converges to a one-per-regime covering assignment which is then pinned. The crisis specialist holds its niche by identity. During dormancy it receives *zero* foreign updates: its decision heads and its stored episode library are structurally untouchable.
2. **Episode-invariant niche training** (solves within-regime drift). Each specialist trains its niche’s decision head on the variance *plus* mean of an SPO-style regret surrogate across the stored episodes of its own regime. Loading on episode-specific regularities inflates the across-episode variance and is penalized away; what survives is the structure every episode shares—which is precisely what the *next* episode will reward.

The framework’s headline empirical object is a 2×2: allocation mechanism (reward-driven vs. reward-independent) crossed with training objective (ERM vs. invariant), evaluated by *post-reactivation decision regret*—mean regret in the first fifteen trading days after a dormant regime returns. The companion paper’s experiments confirm monotone improvement along both axes *and* the signature interaction that makes the two-mechanism account falsifiable: the invariance objective does **nothing** inside a reward-driven monolith ($p = 0.88$ against ERM), because the head it trained was evicted during dormancy; it pays only inside a retaining allocation (-4.4% , $p = 6.1 \times 10^{-3}$). Each mechanism is the other’s precondition.

This document is the explanatory companion to the research paper. It develops the architecture component by component (Section 3), walks through the two non-stationarities with worked numerical demonstrations (Sections 4–6), states the reward-independence principle and the regret decomposition at an accessible level of rigor (Section 8), surveys the experimental evidence including every result that came out against us (Section 9), and maps the application terrain with explicit evidence tiers and an honest “when not to use it” (Section 10).

Reading guide. Readers wanting the mechanism: Sections 3–6. Readers wanting the theory: Section 8 here, full proofs in the mathematical deep-dive companion. Readers evaluating deployment: Sections 9 and 10, especially 10.5. Readers auditing our honesty: every subsection labeled “what did not survive.”

2 Related Work

We position RISP along five lines, ordered by proximity.

GAUSE and reward-independent specialization. GAUSE [1] established, across six real prediction domains and controlled non-stationarity protocols, that retention of dormant regimes under bounded capacity tracks a single property—reward-independence of the capacity assignment plus persistent coverage—and that winner-take-all competition reaches that assignment emergently, matching an oracle fixed assignment without being given it. RISP is the decision-focused, financial instantiation: quality is decision regret rather than prediction reward, competition is batched into multi-day windows, and—the new axis—retained specialists are trained for episode invariance. Everything GAUSE proved about its own scope (the SI-organization audit, the task-incremental framing, the stale-specialist liability under drift) is inherited and re-audited here in the decision setting.

Decision-focused learning. The smart predict-then-optimize framework [5], its surrogates for hard combinatorial layers [6, 7], and the field survey [8] establish why decision regret, not prediction error, is the right training and evaluation signal when forecasts feed an optimizer. All assume stationarity or one-shot shift. Inv-PnCO [4] adds invariant learning across environments for predict-and-optimize, bounding divergences between solution distributions. We adapt its environment-indexed objective to *recurring* regimes (episodes as environments) and add the allocation axis it has no occasion to consider; our Pinsker-type lemma converts its divergence-style guarantees into the regret currency our bound is stated in.

Regime-switching finance. From Markov-switching models [9] through regime-aware allocation [10, 11, 12, 13], this literature detects regimes and conditions a single model. Conditioning answers “what is true now”; it does not answer “who keeps what is true about *then*.” Our capacity-bounded regime-conditioned monolith is this literature’s best case under a memory budget, and the dissociation experiments measure precisely what conditioning alone forfeits.

Online learning over experts. Hedge and multiplicative weights [14, 15], universal portfolios [16, 17], and sleeping experts / adaptive regret [18, 19, 20] are the closest formal relatives and the most dangerous deflation: “is this not just sleeping experts?” No—and the difference is load-bearing. Sleeping-experts theory combines *fixed* experts, which retain dormant state because there is nothing in them to forget. When experts are capacity-bounded *learners*, the combination weights are reward-driven and training resources follow the weights, which reintroduces interference. We run both readings as baselines: Hedge over fixed strategies retains and loses everywhere (it cannot adapt within-regime); Hedge over learning experts adapts and forgets (it tracks the monolith). The contribution lives in the bracket between them.

Continual learning. Catastrophic forgetting [21, 22] and its single-model remedies (EWC [23], progressive networks [24]) treat retention as a regularization or architecture problem. MoE continual-learning theory [25] shows expert isolation mitigates forgetting provided the gate’s updates eventually stop—in our vocabulary, provided the assignment becomes reward-independent. The population mechanism reaches that state with no schedule; the financial contribution of this work is what must be *added* to retention (invariance) once the environment’s recurrences are themselves heterogeneous.

3 The RISP Framework: Architecture for Two Non-Stationarities

3.1 Design Philosophy: Separate the Two Clocks

Financial non-stationarity runs on two clocks. The *fast clock* is regime switching: every few weeks or months the market’s character changes —trend to range, calm to stressed—and the change is largely a *return* to a previously seen state. The *slow clock* is within-regime evolution: each return differs from the last, because the market’s microstructure, participants, and transmission channels drift between occurrences. A system built only for the fast clock (retain every regime’s specialist) arrives at the next crisis perfectly calibrated to the previous one. A system built only for the slow clock (one robust invariant model) averages away

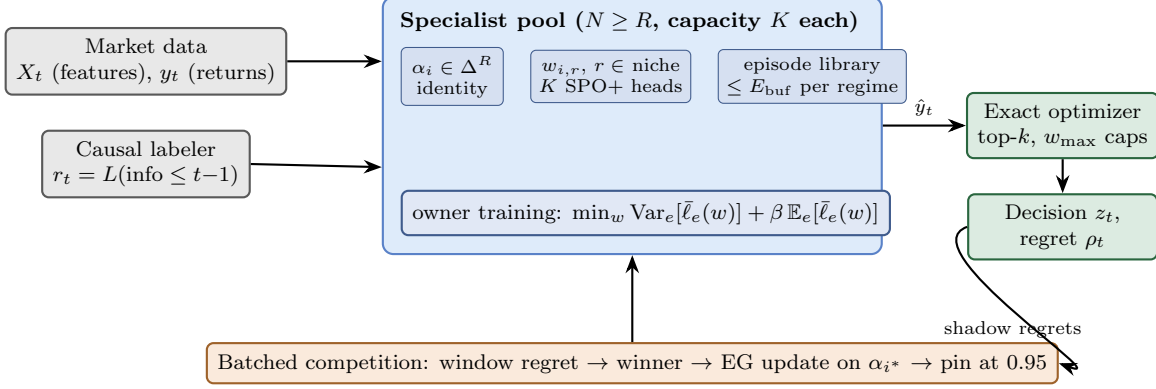


Figure 1 RISP architecture. Serving path (top): label $\rightarrow$ owner’s head $\rightarrow$ exact decision. Learning path (bottom): every specialist’s shadow regret feeds the batched competition, whose winner—and only the winner—updates identity and competence. After pinning, the competition loop is bypassed for that regime and the owner trains alone: allocation has become reward-independent.

the regime structure entirely. The design philosophy of RISP is that the two clocks demand two separate mechanisms coupled through one object—the *specialist*:

The RISP recipe. One specialist per regime, discovered by competition, held by identity, trained for invariance:

1. *Compete*: all specialists shadow-trade every day; cumulative decision regret over a W_c -day window within the active regime selects a window winner.
2. *Specialize*: the winner reinforces its niche affinity for the active regime (exponentiated-gradient on the regime simplex) and trains on the window’s data.
3. *Pin*: when an affinity crosses 0.95, the regime is pinned to its owner; allocation stops responding to reward—the retention guarantee becomes structural.
4. *Generalize*: owners train their niche’s decision head on $\text{Var}_e[\text{regret surrogate}] + \beta \mathbb{E}_e[\text{regret surrogate}]$ across the stored episodes e of their regime—never on one episode alone.

Three deliberate omissions define the architecture as much as its components. There is **no gate**: routing is by the converged assignment, so there is no reward-trained module whose blindness to dormant regimes could erode the inventory. There is **no freezing schedule**: pinning is triggered by the affinity dynamics themselves, not by a hand-set epoch. And there is **no hand-designed regime-to-desk map**: the assignment is discovered, which matters less because design is hard (a fund can silo desks by hand) than because the discovered assignment provably matches the hand-designed skyline while degrading gracefully when the regime inventory is misjudged.

3.2 Architectural Blueprint

Figure 1 shows the data path. Each trading day, a causal labeler maps public information through $t-1$ to a regime label r_t (volatility band $\times$ trend sign in the headline configuration; a filtered two-state model as the robustness alternative). Each specialist i holds: a *niche affinity* $\alpha_i \in \Delta^R$ (its identity—where it belongs); up to K *decision heads* $w_{i,r} \in \mathbb{R}^d$ (its competence—linear SPO+ predictors per held regime); and a bounded *episode library* per held regime (its memory—the raw material for the invariance objective). The serving path is regime label $\rightarrow$ owner $\rightarrow$ head $\rightarrow$ predicted returns $\rightarrow$ exact portfolio optimizer $\rightarrow$ decision. The learning path adds: regret of every specialist’s shadow decision $\rightarrow$ window accumulator $\rightarrow$ winner $\rightarrow$ EG affinity update + invariant training step.

3.3 One Step of the Loop

Concretely, a day in the converged system: the labeler reports **vol-down** (the crisis-type regime). The pinned owner, specialist 3, loads its **vol-down** head—untouched for 280 days but neither evicted nor decayed, because specialist 3 has won nothing and been assigned nothing since its regime went dormant—and its episode library holding the previous four **vol-down** episodes. It predicts cross-sectional returns, the optimizer selects the top- k caps-respecting portfolio, and the realized regret is recorded. The owner then takes one training step—*not* on today’s data alone, but on the variance-plus-mean objective across its stored episodes including the new one: today’s data enters as the newest episode, and the objective immediately punishes any weight that helps today’s episode at the expense of the stored ones. Meanwhile specialists 0–2 served and trained their own active or recent niches; nobody touched specialist 3’s other slots. Tomorrow, if the regime persists, the same; if it ends, specialist 3 goes back to idling—with a library now one episode richer.

4 Decision-Focused Learning: Why Regret Is the Currency

4.1 The Predict-Then-Optimize Gap

The decision layer is a cardinality-constrained long-only portfolio: pick at most k of n assets at weight up to $w_{\max}$ to maximize $y^\top z$. Because the objective is linear, the optimum is exact and instantaneous (the k largest positive predicted returns at full weight), and the *decision regret* of a forecast $\hat{y}$ against realized y is

$$\rho(\hat{y}; y) = \max_{z \in Z} y^\top z - y^\top z(\hat{y}) \geq 0,$$

which is zero iff the forecast selects an optimal portfolio—regardless of how wrong the forecast is numerically.

Observation 1 (Prediction error is the wrong currency). *Two forecasts with identical mean-squared error can differ arbitrarily in regret, and vice versa. A forecast that doubles every true return has large MSE and zero regret (rankings preserved); a forecast with tiny MSE concentrated on swapping the k -th and $(k+1)$ -th assets pays regret on every such day. Training and evaluating specialists by prediction error therefore optimizes—and audits—the wrong object. RISP uses the SPO+ convex surrogate [5] of ρ for training (two oracle calls per gradient) and raw ρ for all evaluation and competition.*

This choice has an audit consequence worth flagging early: GAUSE’s oracle diagnostic found that crypto and commodity panels carry no exploitable regime structure under *prediction* metrics [1]. Since prediction-optimal and decision-optimal differ, that verdict does not automatically transfer—structure invisible to MSE could in principle matter for top- k selection. Re-running the diagnostic under the decision metric is therefore a genuine question, which our E0 answers (Section 9.4): the null transfers. Honest, and not a foregone conclusion.

4.2 Worked Demonstration: When Retention Decides

A miniature with $n = 4$ assets, $k = 1$, $w_{\max} = 1$. The crisis regime’s invariant structure says “asset A (quality) outperforms; asset D (leveraged) crashes,” with true crisis-day returns $y = (+2\%, 0\%, 0\%, -4\%)$.

Retained specialist (held through 280 days of dormancy): predicts $\hat{y} = (+1.5\%, +0.2\%, -0.1\%, -2.8\%)$ —imperfect, but the ranking is right. Decision: A. Regret = 0.

Relearning monolith (evicted the crisis head during dormancy; two days into refitting from prior): predicts noise around zero, $\hat{y} = (+0.1\%, +0.4\%, +0.3\%, +0.2\%)$. Decision: B. Regret = 2% of capital *on that day*; integrated over a fifteen-day probe window at crisis volatility, the relearning transient is the dominant cost in the entire evaluation—which is exactly what the companion paper’s reactivation-aligned traces show at full scale (monolith regret spikes $\sim 40\%$ above its steady state in the probe window, while the retained owner is flat from day one).

Retained but episode-overfit specialist: predicts from the *last* crisis’s accident—say, in that episode, asset C’s sector happened to rally— $\hat{y} = (+1.2\%, 0\%, +1.8\%, -2.5\%)$. Decision: C. Regret = 2%. Retention preserved the head; ERM made the head worth preserving less. This is the second axis, and no allocator fixes it.

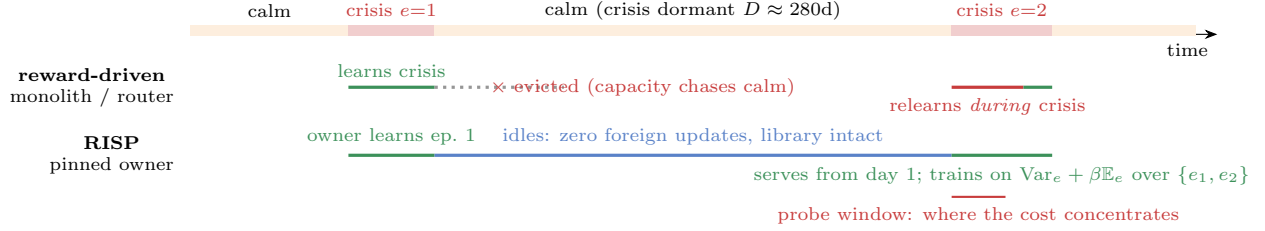


Figure 2 The two failure modes on one timeline. **Top lane:** a reward-driven allocator learns the first crisis, evicts it during dormancy (no reward $\Rightarrow$ no protection), and pays the relearning transient inside the second crisis’s probe window. **Bottom lane:** the pinned owner idles through dormancy structurally untouched—and its invariance objective across episodes $\{e_1, e_2, \dots\}$ keeps what it retains general rather than calibrated to e_1 ’s accidents.

5 The Two Non-Stationarities, Visualized

Figure 2 compresses the paper’s thesis into one timeline: a crisis regime that activates twice, with a long calm dormancy between, drawn for the reward-driven monolith (top) and RISP (bottom).

The figure also locates the two ways to fail *after* solving retention, both measured in the companion paper’s audits: if episodes are heterogeneous and training is ERM, the bottom lane serves episode-1 calibration to episode 2 (the invariance axis, worth 4.4% of probe regret at baseline heterogeneity and growing with it); and if within-regime signal is strong enough to refit in days, the top lane’s relearning transient shrinks until eviction stops mattering—at which point chasing each episode *beats* invariant retention (the E6 inversion, up to -49% at $8\times$ signal-to-noise). The mechanism has boundaries, and both are drawn by experiment rather than assumption.

6 Competition, Affinity, and Pinning

6.1 The Exponentiated-Gradient Affinity Update

Each specialist’s identity is a distribution $\alpha_i \in \Delta^R$ over regimes, initialized uniform. When specialist i^* wins a competition window in regime r , it applies the canonical exponentiated-gradient (Hedge / multiplicative-weights) update [26, 14, 15]:

$$\alpha_{i^*,r} \leftarrow \frac{\alpha_{i^*,r} e^\eta}{\sum_q \alpha_{i^*,q} e^{\eta \mathbf{1}_{[q=r]}}}, \quad \eta = 0.6.$$

The update is multiplicative and simplex-preserving by construction (no clamps, no renormalization pathologies—the lesson of GAUSE’s V3-to-V4 revision is inherited by adopting the canonical form from the start). Six to nine net wins concentrate an affinity from 0.25 to the 0.95 pin threshold.

6.2 Worked Demonstration: From Uniform to Pinned

Four specialists, four regimes, $\eta = 0.6$. Specialist 2 wins regime 0’s first three windows (its early random edge in regime 0 compounds: winners train, training improves, improvement wins):

$$\alpha_2 : (0.25, 0.25, 0.25, 0.25) \xrightarrow{\text{win } r_0} (0.378, 0.207, 0.207, 0.207) \xrightarrow{\text{win } r_0} (0.525, 0.158, 0.158, 0.158) \xrightarrow{\text{win } r_0} (0.668, 0.111, 0.111, 0.111)$$

and five more wins take it past 0.95: pinned. Crucially, specialist 2’s wins *remove* regime 0 from effective contention—other specialists stop winning there (it trains, they don’t), so their affinity mass concentrates on the remaining regimes: competitive exclusion as an assignment algorithm. In every one of the twenty headline seeds this process converges to a *covering, one-per-regime* assignment with distinct owners—and we report its one honest wrinkle: the rare crisis regime, with only a handful of competition windows per run, typically tops out at affinity 0.67–0.87, short of the formal pin threshold, while still having a unique stable owner who serves it throughout. Scarce regimes converge in *ownership* before they converge in *certainty*; deployments should pin on ownership stability, not on the affinity scalar alone.

6.3 Batched Windows: A Prediction the Data Revised

GAUSE competes every step; we pre-registered the expectation that per-step competition would fail in markets (daily regret differences are noise-dominated; a single-window Hoeffding bound says the wrong winner is selected with probability $\approx \frac{1}{2} - O(\Delta/\sigma_n)$) and made the W_c -day batched window the headline financial adaptation. The ablation refused the story: post-reactivation regret is flat across $W_c \in \{1, 5, 20, 60\}$ (0.0303–0.0309), with $W_c = 1$ marginally *best*. The resolution is instructive: the assignment is decided not by one window but by the *accumulation* of EG updates across hundreds of windows, and that accumulation is itself an averager—noisy single-window winners wash out, persistent quality differentials compound. The single-window analysis was correct about the question it answered and irrelevant to the question that mattered. We keep $W_c = 20$ as the default (one auditable affinity event per fortnight is operationally attractive) and report the revision as what it is: a pre-registered mechanism story, falsified by its own ablation, replaced by a better mechanism story—and the framework is robust either way.

6.4 Pinning: Where Reward-Independence Becomes Structural

Before pinning, the assignment is still reward-*driven* (wins are reward events). The retention guarantee dates from the pin: from that moment, regime- r data routes to its owner regardless of anyone’s recent performance, and the owner receives no foreign assignments regardless of its own. The never-pin ablation quantifies what perpetual competition costs (0.0320 vs. 0.0307): an unpinned population re-admits a thin channel of reward-coupling—a lucky streak by a non-owner during a regime block diverts that block’s training—and pays for it at reactivations, though the EG dynamics’ inertia keeps the damage far below router levels. Pinning is the moment the mechanism crosses the line that Proposition 1 (below) draws between the forgetting and retaining classes.

7 Bounded Capacity, Memory Models, and the Eleven Arms

7.1 Two Readings of “Capacity”

Every specialist holds at most $K < R$ regime heads, under two memory models. The **hard (LRU)** model makes capacity slot-like: learning an unheld regime at full capacity evicts the least-recently-used head *and its episode library*—knowledge lost is data lost, modeling playbooks that must be actively maintained to exist. The **soft (interference)** model has no eviction: every foreign update decays all other held heads by a factor $(1 - \rho_{\text{int}})$, modeling parameter interference in shared representations. The companion paper’s capacity sweep shows the dissociation under both, with one asymmetry worth knowing: under hard slots the reward-driven arms close the gap fully at $K = R$ (no eviction pressure, nothing to protect), while under interference they never quite close it (0.0328 vs. 0.0307 at $K = R$)—shared representations leak even without scarcity, which is the realistic reading for parametric models.

7.2 The Arm Inventory

The eleven arms share identical specialists, learning rates, buffers, and data; they differ only in *who learns what when* and *which loss*. Reward-driven allocations: **A1** capacity- K monolith (always trains the active regime—regime-conditioned finance under a capacity budget); **A2** MoE router (gate trained on realized regret); **A3** recent-performance capital allocator (softmax of trailing 60-day performance; training intensity follows capital—the industry default, whose dormant-desk erosion rate is set by its softmax floor); **A8b** Hedge over learning experts (training follows the top weight). Reward-independent allocations: **A4** random fixed niches (coverage gaps possible); **A5** RISP-ERM; **A6** RISP-full; **A9/A10** hand-pinned oracle assignment with ERM / invariant training (the skylines). Orthogonal probes: **A7** monolith with the invariant objective (can a loss substitute for an allocator? No—the heads it trains are evicted all the same); **A8a** Hedge over fixed strategies (the sleeping-experts reading: retains trivially, adapts never).

8 The Reward-Independence Principle, Extended by Invariance

8.1 Why a Reward-Driven Allocator Must Forget

Observation 2 (Dormant regimes emit no protective signal). *During regime r ’s dormancy, every signal a reward-driven allocator consumes—realized P&L, regret, gate gradients—is generated by other regimes. The allocator is not merely disinclined to protect r ’s specialist; it is informationally incapable of it: no observable event during dormancy distinguishes “this capacity guards a regime that will return” from “this capacity is waste.” Each dormant step therefore carries positive probability of a foreign assignment to r ’s specialist, and over a dormancy of length D the count of distinct foreign acquisitions U_D grows without bound: hard eviction triggers at $U_D \geq K$ (probability $\rightarrow 1$ exponentially in D), soft knowledge decays as $(1 - \rho_{\text{int}})^{U_D} \rightarrow 0$.*

Principle 1 (Reward-independence, plus coverage, plus invariance). *Within the class of mechanisms whose only protection is the assignment itself: (i) retention of a dormant regime requires the assignment to be independent of current reward and to cover that regime—the GAUSE principle, inherited intact; and (ii) in environments whose regimes return changed, retention is necessary but not sufficient for post-reactivation performance: the retained head must additionally have been trained for across-episode invariance, because what a reward-blind dormancy preserves is whatever the objective put there—structure if the objective demanded structure, episode accidents if it permitted them.*

Proposition 1 (Regret decomposition, informal statement). *Post-reactivation regret is bounded by an invariance gap plus a forgetting deficit: the first is the mean-plus-variance of the specialist’s episode risks (controlled by the $\text{Var}_e + \beta \text{E}_e$ objective; convertible from divergence-style invariance guarantees at the price of one Pinsker step), the second is the eviction probability times the relearning cost (zero for reward-independent covering assignments, saturating to full cost for reward-driven ones). The two terms have disjoint knobs—which is the formal content of the 2×2, and the reason the empirical interaction is super-additive: the invariance knob acts on a head that only the retention knob can keep in existence.*

Full statements and proofs (Cantelli transfer, Hoeffding-over-episodes estimation price, the Pinsker bridge with explicit constant $B = 2kw_{\max}y_{\max}$, the eviction-rate computation, and the break-even-dormancy window) are in the mathematical deep-dive companion; the research paper carries the citable versions.

8.2 The Break-Even Audit: Idle Desks Are Not Free

Retention’s benefit is a one-shot saving per reactivation; its cost is a carrying charge on every dormant day. Charging c per idle specialist-day against the measured relearning saving yields a *window* of dormancy lengths in which retention pays: long enough that the reward-driven alternative would certainly have evicted ($D \gtrsim 40$ –60 days at our calibration), short enough that cumulative carrying cost does not swamp the saving (at 10% relative carrying cost, $D \lesssim 600$ days; above $\approx 15\%$, *no* dormancy pays). A fund whose crisis playbook reactivates once a decade may rationally let it decay; one whose playbook reactivates yearly may not. The framework’s honest deliverable is the formula with measured constants, not a universal recommendation.

9 Experiments and Evaluation

9.1 Protocol

All experiments: 20 seeds, Welch tests with Holm–Bonferroni correction, 95% CIs, identical data streams across arms within a seed. Synthetic markets calibrate signal-to-noise to realistic daily cross-sections ($|\theta| \approx 0.012$ vs. noise 0.02–0.06, crisis noise $3\times$), $n = 20$ assets, $d = 20$ features (two invariant, one episode-spurious, sixteen distractors), $R = 4$ regimes with the crisis regime dormant 250–450 days between episodes; 15 evaluated reactivations per run. The probe window is the first 15 days after a reactivation following ≥ 90 days of dormancy. Three data tiers: synthetic (headline—full experimental control), regime-stitched real crypto (real cross-sections, controlled schedules), and real walk-forward panels (the diagnostic tier). *Every percentage below traces to a released result file.*

9.2 Results on the Headline Dissociation (E1)

Post-reactivation regret, the four cells of the 2×2 : reward-driven ERM (monolith) 0.0342; reward-driven invariant 0.0344 ($p = 0.88$ against the monolith—**the objective does nothing without retention**); reward-independent ERM 0.0321 (-6.1% , $p = 1.3 \times 10^{-3}$ —**the retention axis**); reward-independent invariant 0.0307 (a further -4.4% , $p = 6.1 \times 10^{-3}$ —**the invariance axis**; jointly -10.3% , $p = 9.2 \times 10^{-7}$, 38% of the excess above the irreducible noise floor). The per-seed interaction index is -0.0015 ± 0.0005 : super-additive, CI excluding zero. Around the square: the learned router is statistically the monolith ($p = 0.83$); the recent-performance allocator is *worse* than the monolith (0.0378—its capital floor keeps eroding dormant specialists in steady state too); Hedge-over-fixed retains and still loses everywhere (0.0439; nothing to forget, nothing learned); Hedge-over-learners tracks the monolith (0.0341). And the skyline: RISP-full is statistically indistinguishable from the hand-pinned oracle assignment with the same objective (0.0307 vs. 0.0305, $p = 0.72$), and nominally better than the oracle assignment with ERM ($p = 0.011$)—**discovered assignment plus invariant training beats designed assignment plus ERM, and matches designed assignment plus invariant training without being designed**.

9.3 Results on the Boundaries (E2, E3, E4, E6)

Capacity (E2): RISP arms are flat in K (a converged owner needs one slot); reward-driven arms close the gap only at $K = R$ under hard slots and not even there under interference. **Dormancy (E3):** the retention gap rises from 0.0015 at $D = 21$ to ≈ 0.004 at $D \geq 63$ and saturates—the eviction event, once near-certain, cannot recur. **Heterogeneity (E4):** fixing the retaining allocation and sweeping episode heterogeneity, ERM degrades about 23% faster than the invariant objective, and the ERM–INV gap widens monotonically from 0.0010 to 0.0024 ($2.4\times$)—the invariance margin is purchased against episode variability specifically. **Signal strength (E6):** the audit that began as our own failed prototype. At $0.5\text{--}2\times$ baseline signal-to-noise the RISP advantage is $+7\text{--}10\%$; at $4\times$ it is -13% ; at $8\times$, -49% : with strong, fast-learnable, episode-specific signal, greedy refitting beats invariant retention, and the skyline arm inverts identically (it is the regime, not the mechanism). The crossover near $2\text{--}3\times$ is the deployment boundary, and locating it is part of the contribution.

9.4 What Did Not Survive (E0, E1s, and Two Predictions)

The real-data gate fails, and the protocol works. The pre-registered structure diagnostic—oracle regime-conditioning vs. block-shuffled labels on the *decision* metric, walk-forward, two causal labelers—finds no exploitable regime structure in five-pair daily crypto or three-commodity panels (all $|z| < 0.7$; conditioning = shuffled = pooled). The full eleven-arm experiment on regime-stitched real crypto is consequently *flat* (all pairwise $p > 0.2$)—exactly what the failed gate predicts, demonstrated rather than assumed: the same machinery that separates eleven arms at $p \sim 10^{-6}$ where structure exists produces a null table where it does not. An evaluation that cannot go flat should not be trusted when it goes sharp. **Two pre-registered stories were refuted:** the batching expectation (Section 6; flat in W_c) and —from the development history—the assumption that the mechanism would show at any signal-to-noise (the first prototype’s high-SNR market showed nothing, and became E6). Both revisions are reported in the companion paper at the same prominence as the confirmations.

9.5 Ablations

The invariance objective needs both terms: variance-only ($\beta = 0$) collapses to 0.0356, *worse than retained ERM* (a head can be uniformly bad with zero variance); the plateau $\beta \in [0.25, 4]$ gives 0.0307–0.0317; $\beta = 16$ drifts back to ERM-like behavior. Competition alone works ($\lambda = 0$: 0.0314 vs. 0.0307 with the niche bonus—the bonus accelerates, does not cause). Pinning matters (0.0307 pinned vs. 0.0320 never-pinned). And the entire dissociation is robust to swapping hard eviction for soft interference.

9.6 Relation to Mixture-of-Experts Continual-Learning Theory

Recent MoE theory [25] proves that expert isolation mitigates forgetting in overparameterized settings under two provisos: experts plentiful relative to tasks, and gating updates *terminated* after an exploration phase. Both provisos are restatements, in our vocabulary, of the reward-independence conjunction: terminated gate updates make the assignment reward-independent; plentiful experts make coverage free. The financial setting violates both by default—capacity is scarce and the gate (capital allocation) never stops moving—and our A2/A3 arms measure exactly what the violation costs. RISP’s pinning is gate-freezing with an emergent schedule: the freeze happens per-regime, when that regime’s ownership has stabilized, rather than at a hand-set epoch. The invariance axis has no analogue in the MoE theory because its tasks return unchanged; ours return changed, which is why isolation alone is not enough here.

9.7 Statistical Protocol and Reproducibility

Every experiment is seeded end-to-end: seed s fixes the schedule, the market draw, and each arm’s internal randomness; arms within a seed see identical data (paired design). Headline comparisons are Welch two-sample tests over the 20 per-seed means, Holm–Bonferroni-corrected within each experiment’s pre-registered pair family; all intervals are 95% normal CIs over seeds. We do not quote Sharpe ratios for synthetic results (regret against a known optimum dominates them informationally), and we pre-commit the real-data battery—deflated Sharpe [27], SPA/Reality Check [28, 29], 5/10/25 bps costs, nested walk-forward hyperparameters—for any future panel that passes the E0 gate. The repository layout mirrors the claims: `code/` (mechanism and arms), `results/*.json` (one file per experiment; every number in every table traces to one), `paper/figures/` (generated by one script from those files). Re-running the full battery takes under one hour on a laptop; no GPU is used.

9.8 Headline Numbers at a Glance

Arm (class)	Post-react.	Overall	Steady
A10 oracle + INV (<i>skyline</i>)	0.0305 $\pm$ 0.0008	0.0259	0.0250
A6 RISP-full (<i>ours</i>)	0.0307 $\pm$ 0.0007	0.0260	0.0251
A9 oracle + ERM (<i>skyline</i>)	0.0321 $\pm$ 0.0008	0.0274	0.0262
A5 RISP-ERM	0.0321 $\pm$ 0.0007	0.0275	0.0264
A4 random fixed (ERM)	0.0330 $\pm$ 0.0013	0.0283	0.0272
A2 MoE router	0.0341 $\pm$ 0.0011	0.0270	0.0248
A8b Hedge over learners	0.0341 $\pm$ 0.0010	0.0267	0.0244
A1 monolith (ERM)	0.0342 $\pm$ 0.0010	0.0269	0.0246
A7 monolith (INV)	0.0344 $\pm$ 0.0010	0.0266	0.0243
A3 recent-perf allocator	0.0378 $\pm$ 0.0010	0.0294	0.0269
A8a Hedge over fixed	0.0439 $\pm$ 0.0009	0.0382	0.0370

Table 1 E1 headline (synthetic, $K=2$, hard memory, 20 seeds, mean $\pm$ 95% CI). Reading order: the retaining classes occupy the top block; every reward-chasing learner clusters at the bottom regardless of architecture; the fixed-expert Hedge floor shows what retention without learning is worth.

9.9 Worked Sensitivities

Three further numbers practitioners ask for first. *How long must dormancy be?* The retention gap is 0.0015 at $D = 21$ trading days, 0.0038 at $D = 63$, and saturates near 0.0040–0.0044 by $D = 126$ –504: roughly, dormancy beyond one quarter makes eviction near-certain for a $K = 2$ learner under three active regimes, and further dormancy changes nothing (the eviction event cannot recur). *How sensitive is the invariance margin to β ?* Flat within $[0.25, 4]$ (0.0307–0.0317); the failure modes are at the edges ($\beta = 0$: 0.0356; $\beta = 16$: 0.0352)—tune lazily, but do not drop either term. *How much does discovery cost relative to design?* At convergence, nothing measurable: RISP-full vs. the hand-pinned oracle with the same objective is 0.0307

Arm	Signal-to-noise multiplier (E6)				
	0.5×	1×	2×	4×	8×
A1 monolith post-react.	0.0378	0.0342	0.0299	0.0273	0.0329
A6 RISP post-react.	0.0347	0.0307	0.0278	0.0309	0.0491
RISP advantage	+8.3%	+10.3%	+7.0%	−13.1%	−49.3%

Table 2 The E6 inversion: the mechanism’s advantage by signal strength. Bold cells are the honest boundary—where within-regime signal is strong and fast to refit, retention-plus-invariance loses to greedy episode-chasing, and the hand-pinned skyline inverts identically.

vs. 0.0305 ($p = 0.72$); during the discovery phase (first ~ 15 –25 competition windows per common regime) the population pays an exploration cost visible in the overall metric, which is the price of not knowing the regime inventory in advance.

10 Potential Applications

We tier the application survey by the strength of present evidence, in the GAUSE convention: *validated-mechanism* (the controlled dissociation applies directly), *mechanism-level* (the structure matches but the domain experiment has not been run), and *candidate* (plausible mapping, open).

10.1 Multi-Strategy Capital Allocation (mechanism-level)

The direct reading. Desks are specialists; the regime inventory is the house view; recent-P&L allocation across desks is arm A3—and the experiments say A3 is *worse* than a naive monolith, because its capital floor erodes dormant desks continuously rather than at discrete evictions. The framework’s translation into practice is not “replace your allocator with a competition” but three auditable properties any allocator can be tested for: (i) does a desk’s *maintenance budget*—data, recalibration, staffing—decouple from its trailing P&L once its mandate is pinned? (ii) does every regime in the house inventory have exactly one accountable owner? (iii) are dormant desks’ models retrained on *all* historical episodes of their regime with an across-episode robustness criterion, rather than fine-tuned to the last occurrence? Properties (i)+(ii) are the retention conjuncts; (iii) is the invariance axis; the E1 table prices each. The break-even analysis (carrying cost vs. reactivation saving) gives the CFO-facing arithmetic for whether a given playbook deserves an idle owner at all.

10.2 Risk-Model Committees and Stress Calibration (mechanism-level)

Stress-VaR and crisis-correlation models are the purest dormancy case: their regime activates rarely, the cost of stale calibration concentrates entirely in the activation window, and—unlike trading desks—their “P&L” during calm periods is exactly zero, making them maximally vulnerable to any reward-coupled resource allocation (model-validation attention follows recent usage). The RISP reading: pin ownership of each stress scenario family; retain its episode library (1998, 2008, 2011, 2020, 2022 as environments); calibrate by across-episode invariance rather than to the most recent stress event. Dormancy here is measured in years, relearning cost in regulatory capital and mis-hedged tails. The roadmap’s fallback plan F3 designates this the pivot application if trading-side dormancy proves too short to matter; the present experiments make it mechanism-level, not validated—the stress-calibration analogue of E1 has not been run.

10.3 Cross-Asset and Cross-Frequency Extensions (candidate)

The episode-scarcity mitigation—counting the same crisis across asset classes as distinct environments—upgrades the invariance claim from within-asset to *cross-asset* invariance of crisis structure, which is both stronger and more useful (a crisis specialist whose decision structure transfers across equities, credit, and FX). Intraday frequencies multiply episode counts and shrink dormancy; whether regime structure passes the

E0 gate at those frequencies is an empirical question our protocol makes cheap to ask. Both are candidates: the data work is real and unrun.

10.4 Beyond Finance (candidate)

Any domain with recurring-but-drifting modes and capacity-bounded learners fits the template: epidemic-wave forecasting (each wave a new episode of a recurring regime), seasonal demand systems, climate-regime-conditioned prediction, adversarial security postures (attack families recur, mutated). The GAUSE companion documents the retention half across six non-financial domains; the invariance half should transfer wherever episodes are heterogeneous draws of a stable structure. We flag these as candidates and resist further speculation.

10.5 When Not to Use It

The audits draw five boundaries, each measured rather than assumed:

1. **No verifiable regime structure** (E0 fails): the machinery will organize anyway—organization is not structure—and retain nothing of value. Run the diagnostic first; ours failed on daily crypto and commodities, and we shipped the null.
2. **Strong, fast-learnable within-regime signal** (the E6 inversion): if a fresh model refits in days and episode-specific signal is worth chasing, greedy ERM beats invariant retention by up to 49%. High-frequency settings with rich cross-sections likely live here.
3. **No real dormancy** (E3 at small D): regimes that revisit every few weeks never trigger eviction; the retention gap at $D = 21$ is a quarter of its asymptote.
4. **Slack capacity with hard isolation** (E2 at $K = R$): if every model can hold every regime without interference, conditioning suffices. (Under shared representations the gap never fully closes—check which reading describes your stack.)
5. **Carrying costs above break-even**: an idle specialist whose regime returns once a decade may not be worth its upkeep; the Proposition-level arithmetic with your own c , D , and relearning cost decides, and “let it decay, plan to relearn” is sometimes the rational answer.

11 Conclusion and Future Work

RISP is a two-mechanism answer to a two-clock problem. Reward-independent capacity assignment—discovered by competition, fixed by pinning—makes dormant expertise structurally untouchable, zeroing the forgetting deficit that every reward-chasing allocator provably accumulates. Episode-invariant training makes the untouchable expertise worth touching again, closing the within-regime generalization gap that retention alone preserves intact. The two axes are experimentally separable, individually significant, super-additive in combination, and jointly sufficient to match a hand-designed oracle assignment that the system was never shown. Around the mechanism we built the evaluation discipline we want to be held to: pre-registered orderings (two of which the data refused, both reported), a structure diagnostic that runs before claims (and failed on our real panels, a null we ship at headline prominence), and audits that convert every assumption into a measured boundary—capacity, dormancy, heterogeneity, signal strength, carrying cost.

Future work, in order of leverage: (i) the **equities flagship**—a survivorship-aware S&P 500 constituents panel, 1990–2025, whose 2000/2008/2011/2015/2018/2020/2022 stress episodes supply the 5–7 crisis environments the invariance objective wants, with the E0 gate as the pre-registered go/no-go; (ii) **cross-asset episodes**, upgrading within-regime invariance to cross-asset crisis structure; (iii) a **staleness trigger** in the GAUSE style, re-opening competition on confident underperformance, to soften the E6 inversion where within-regime drift is exploitable; (iv) **class-incremental operation**—inferring the regime while specializing in it—which GAUSE showed collapses under real feature overlap and which the financial setting will not make easier; and (v) the **risk-committee deployment**, where dormancy is longest, reward-coupling

most insidious, and the retention argument needs no P&L to state. The wager of this research program is unchanged from its companion: in non-stationary worlds, *who is allowed to keep knowing things*—and *what they are trained to keep knowing*—matter more than how well anything is known today.

A Appendix: Arm Implementation Details

All arms instantiate the same **Specialist** class: linear heads $w \in \mathbb{R}^{20}$ trained by SGD on the SPO+ surrogate (learning rate 0.08, two steps per observation, minibatches of 8 stored days), episode buffers of at most 6 episodes $\times$ 40 days per held regime, capacity K heads with hard-LRU eviction (head and buffer reset) or soft interference (foreign updates decay other heads by $\rho_{\text{int}} = 0.06$). Differences are confined to allocation:

A1/A7 monolith. One specialist; every observation routes to it; it (re)creates the active regime’s head on demand, evicting LRU. A7 differs only in training loss (Eq. of Section 3, recipe step 4).

A2 MoE router. $N = 4$ experts, gate matrix $G \in \mathbb{R}^{R \times N}$. Serving routes regime r to $\arg \max_j G[r, j]$; training routes with $\epsilon = 0.1$ exploration; the routed expert learns; the gate updates $G[r, j] += 0.25 (b_r - \rho_j)$ with b_r an EMA regret baseline per regime—reward-driven by construction, and silent for any dormant r .

A3 recent-performance allocator. Capital shares $v = \text{softmax}(\text{trailing 60-day mean negative regret}/0.02)$; serving follows $\arg \max v$; *every* specialist trains on the active regime with probability v_i per day. The softmax floor ($\sim 5\text{--}10\%$ for losers) is the erosion rate of dormant expertise—and the lever a fund could pull to make A3 partially reward-independent (a hard capital floor with mandate protection is a pin by another name).

A4/A9/A10 fixed assignments. A4: each specialist draws K random niches at initialization; first claimer owns; unowned regimes are served uninformed (the coverage gap). A9/A10: the identity permutation (specialist i owns regime i)—reward-independent, perfectly covering, zero discovery cost; the legitimate skyline.

A5/A6 RISP. As Algorithm 1 of the companion paper: shadow-evaluate all, accumulate window regret, window winner does one EG affinity update ($\eta = 0.6$) and trains on the window’s days; pin at affinity 0.95; after pinning, owner-only serving and training. A5 trains ERM; A6 trains $\text{Var}_e + \beta \mathbb{E}_e$ with $\beta = 1$.

A8a/A8b Hedge. A8a: eight fixed linear strategies ($\pm$ each non-constant feature direction), exponential weights with $\eta_H = 0.1$ on daily regret, serve the top weight. A8b: the same weighting over four capacity- K *learning* specialists; the top-weight specialist trains on the active regime each day.

B Appendix: Synthetic Market and Schedule Parameters

C Appendix: The E0 Diagnostic, Step by Step

The gate any real deployment should run first, in five steps: (1) construct causal regime labels $r_t = L(\text{info} \leq t-1)$ —we use volatility-percentile band $\times$ trend sign (L1) and a forward-only filtered two-state volatility model (L2); (2) fit one ridge predictor per regime, walk-forward (refit every 60 days on all past data of that regime); (3) score by *decision regret* on the held-out second half; (4) repeat with block-shuffled labels (permute each contiguous regime block’s identity; 10 draws)—this preserves block structure and destroys label-structure alignment; (5) compare. Pass = real labels beat shuffled at conventional significance. Our panels: crypto L1 gap -1.5% ($z = -0.66$), crypto L2 -0.7% , commodities L1 -0.3% , commodities L2 $+0.2\%$ ($z = 0.16$): **fail**, uniformly—and the pooled unconditioned model matches both, i.e., regime conditioning

Parameter	Value	Role
n assets	20	cross-section width
d features	20	2 invariant + 1 spurious + 16 distractors + const
$k, w_{\max}$	4, 0.25	decision layer (top- k at caps)
R regimes	4	trend / range / high-vol / crisis
$ \theta_r $	0.012	invariant signal scale (daily)
$\gamma_{r,e}$ scale	0.012 $\times$ het	episode-spurious loading, redrawn per episode
noise	$0.02 \times (1, 0.75, 2, 3)$	per-regime daily idiosyncratic vol
blocks	25–60d common; 15–30d crisis	regime persistence
crisis cadence	every 4th cycle	dormancy 250–450d
T	$\approx 2,370$ d	first half burn-in, second half evaluated
probe	15d after ≥ 90 d dormancy	post-reactivation window
N, K	4, 2 (swept 1–4)	population and capacity
$W_c, \eta, \lambda, \text{pin}$	20d, 0.6, 0.3, 0.95	competition (each ablated)
β	1.0 (swept 0–16)	invariance weight
seeds	20	all CIs and tests

Table 3 Default configuration. Every row marked “swept” or “ablated” has a dedicated experiment in the companion paper.

buys nothing on the decision metric here. The honest scope of this null: five crypto pairs and three commodities, daily bars, this feature inventory, these two labelers. The protocol’s value is that the same five steps, run on a richer panel that passes, would activate the full E1 battery with pre-registered predictions already on file.

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